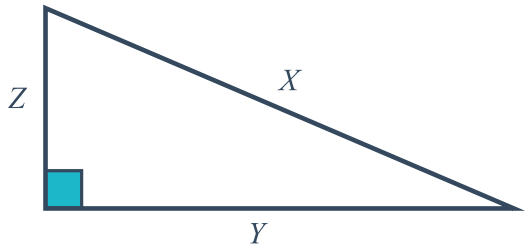
The Pythagorean theorem and its converse

1 Identify which side of the given triangle is the hypotenuse.



2 If 3 and 4 are the two smallest values in a Pythagorean triple, what is the largest value, c, in the triple?

3 Using your knowledge of common Pythagorean triples , find the hypotenuse in a triangle with shorter side lengths 15 units and 36 units.

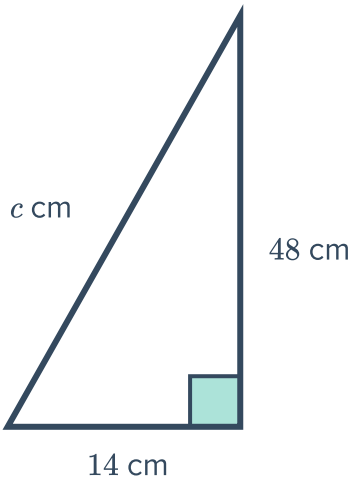
a This triple is a multiple of a more common Pythagorean triple. Which one?

b What is the length of the unknown side in the triangle?

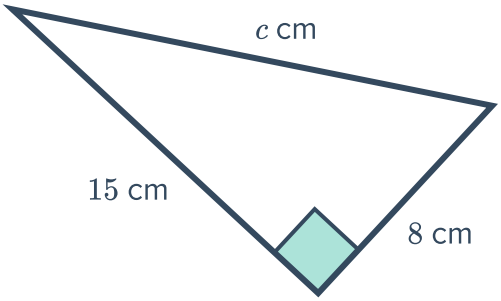
4 If x, y, and z are three sides of a right triangle, will side lengths of $4x$, $4y$ and $4z$ make a right triangle as well?

5 Calculate the value of c in the following triangles, rounding your answers to two decimal places:

a

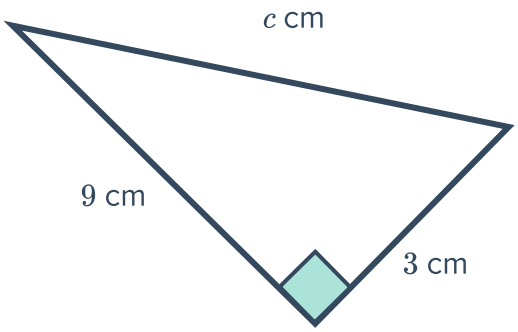


b



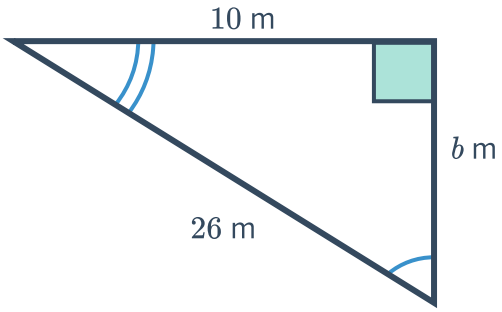
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c

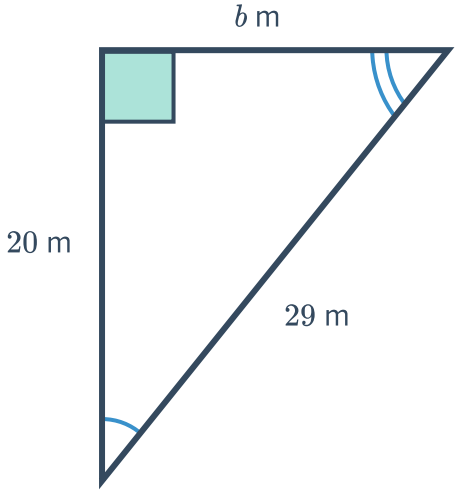


6 Calculate the value of b in the following triangles, rounding your answers to two decimal places:

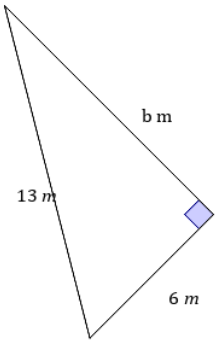
a



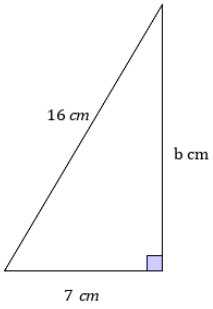
b



c

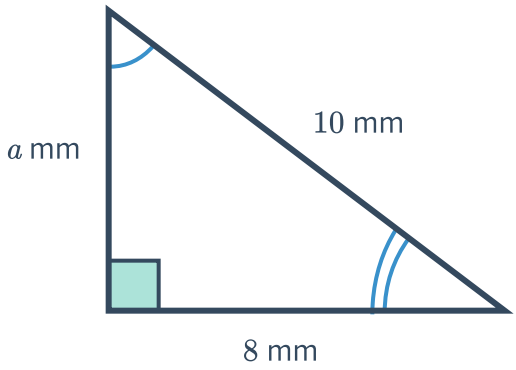


d

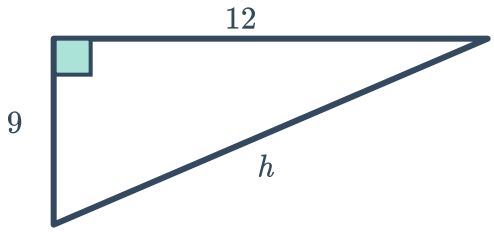


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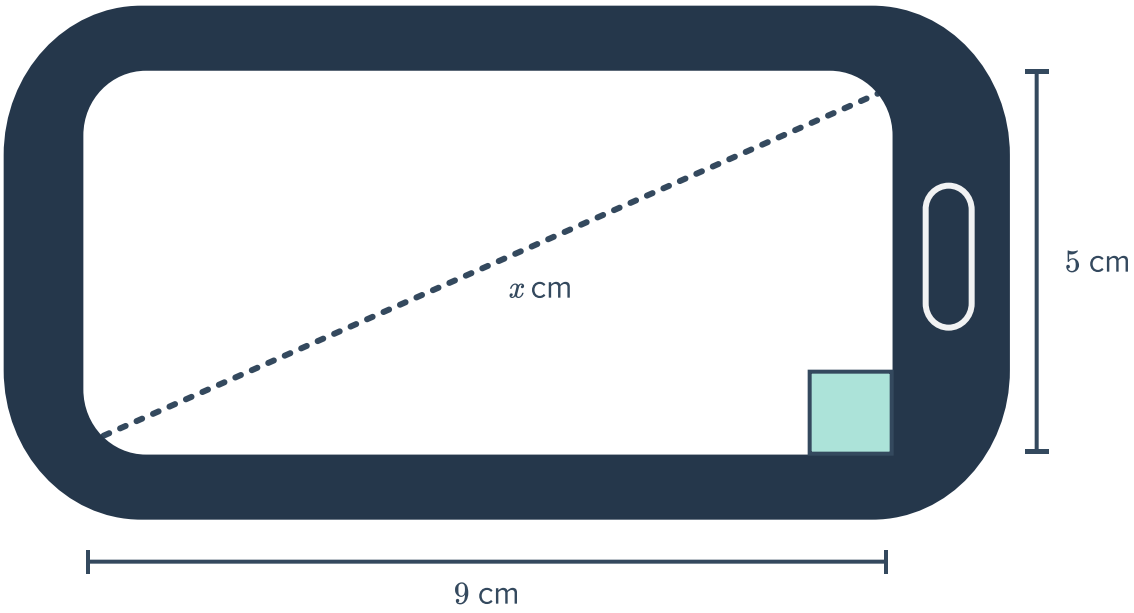
7 Calculate the value of a in the following triangle:



8 Iain's car has run out of gas. He walks 12 km west and then 9 km south looking for a gas station.
If he is now h km directly from his starting point, find the value of h.



9 The screen on a handheld device has dimensions 9 cm by 5 cm, and a diagonal of length x cm.
Find the value of x, correct to two decimal places.



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10 Find b, where b mm is the length of one side of a right triangle whose hypotenuse is 5 mm in length and whose other side is 4 mm in length. Round your answer to two decimal places.

11 A right isosceles triangle has two sides measuring 17 cm.

a Find the length of its hypotenuse, c, rounding your answers to two decimal places.

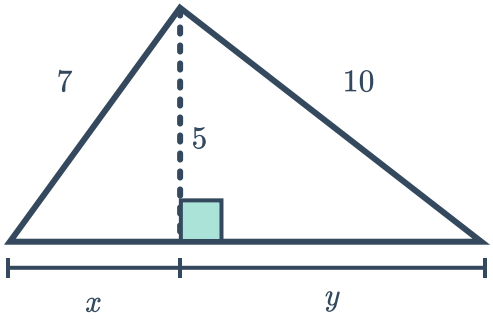
b Calculate the perimeter of the triangle, rounding your answers to two decimal places.

12 Consider the given triangle:
Find the following, rounding your answer to two decimal places:

a The value of x.

b The value of y.

c The length of the base of the triangle.



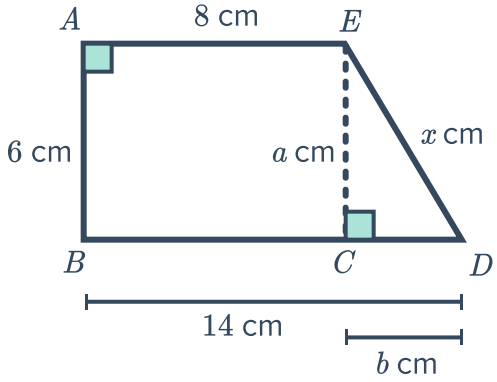
13 Consider the following trapezoid:

a Find the value of a.

b Find the value of b.

c Find x, correct to two decimal places.

d Find the perimeter of the trapezoid, correct to two decimal places.



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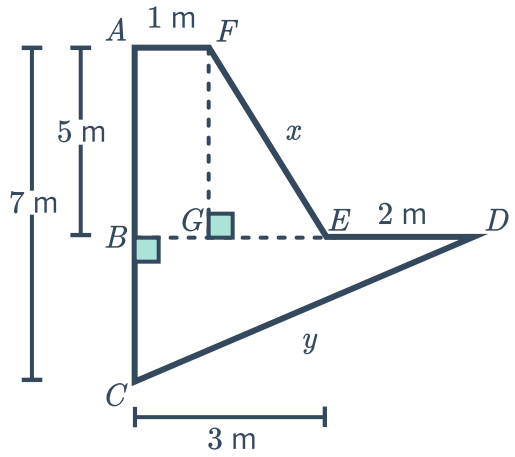
14 A farmer wants to build a fence around the entire perimeter of his land, as shown in the diagram below. The fencing costs $\$37$ per meter.

a Find the value of x. Round your answer to two decimal places.

b Find the value of y. Round your answer to two decimal places.

c How many meter of fencing does the farmer require, if fencing is sold by the meter?

d At $\$37$ per metre of fencing, how much will it cost him to build the fence along the entire perimeter of the land?

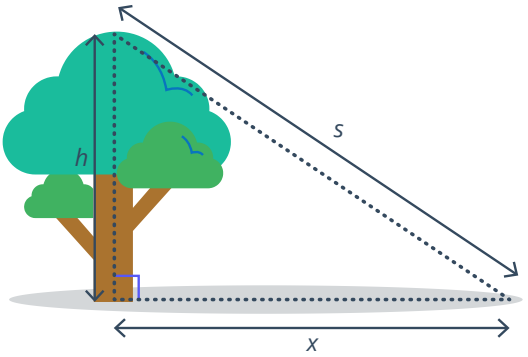


15 Consider the following triangle:

a Name the hypotenuse.

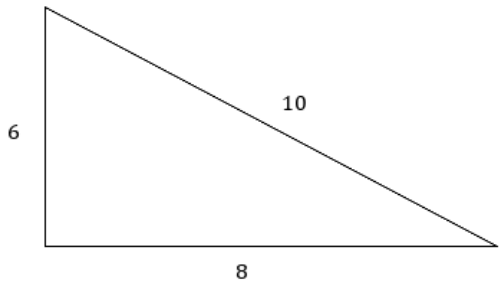
b Write the Pythagorean theorem for this triangle.

c Is $x + h > s$?

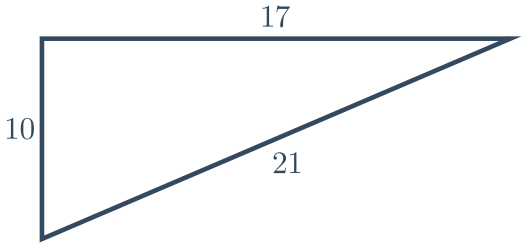


16 Use the Pythagorean theorem to determine whether the following triangles are right-angled:

a



b



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17 Consider a triangle whose shortest sides have lengths 9 and 12. The longest side of the triangle has a length of c.

a Find the value of c, if the triangle has a right angle.

b State whether the triangle is right, acute, or obtuse if:

i $c = 17$ **ii** $c = 14$

18 Consider three straight lines with lengths 8, 6 and 19 units.

a Is it possible to form a triangle using these lines? Explain your answer.

b How could the side of length 19 units be changed so that a triangle can be made?

19 For each of the three straight lines with given lengths:

i Is it possible to form a triangle using these lines? If yes, answer part (ii).

ii Is the triangle an acute, obtuse, or right triangle? Justify your answer mathematically.

a 12, 9 and 13 units

b 8, 6 and 11 units

c 8, 15 and 17 units

d 6, 4 and 7 units

e 9, 15 and $\left(3\sqrt{34}+5\right)$ units

f 7, 5.1 and 8 units

g 4.9, 13.2 and 16.1 units

h 4, $\sqrt{20}$ and 5 units

i $\sqrt{131}$, $\sqrt{337}$ and 25 units.

j $\sqrt{13}$, $\sqrt{19}$ and $\sqrt{32}$ units

20 The sides of a triangle have lengths (in descending order) of $3x$, $2x$ and $\sqrt{15}$. For what values of x is the triangle obtuse?

21 For the given lengths of the sides of a triangle in descending order, determine the values of x that will make the following triangles acute.

a 12, $5x$ and $\sqrt{7}x$

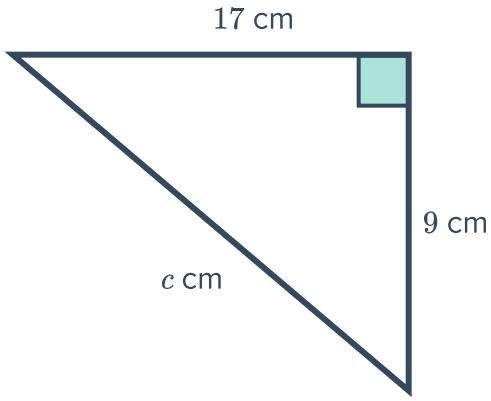
b $2x$, 12 and 8

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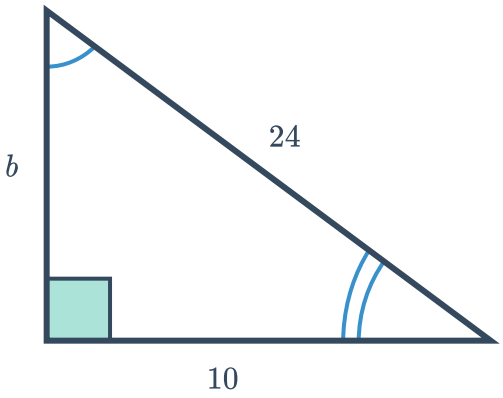
Additional questions

22 Find the value of the variable in the following triangles. Round your answer to two decimal places.

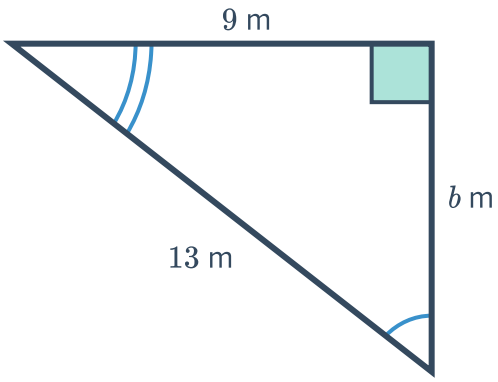
a



b



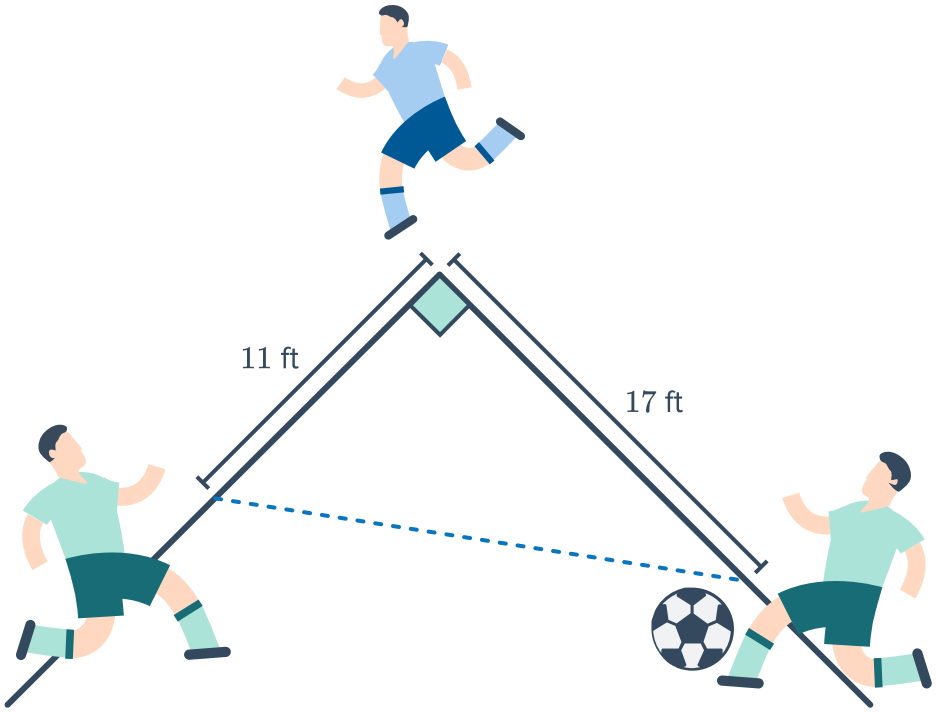
c



23 Andriana knows the two largest numbers in a Pythagorean triple are 41 and 40. Determine the number which Andriana needs to complete the triad.

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24 Fred and Carlo are playing soccer together. An oppostion player who is 11 ft away is about to tackle Fred so he passes the ball to Carlo who is 17 ft away from the opposition player. If Fred and Carlo's lines of sight to the opposition player are perpendicular, find the distance the ball travels. Round your answer to two decimal places.



25 Find the length of the unknown side, b, in the following right triangles. Round your answer to two decimal places:

a A triangle whose hypotenuse is 3 ft in length and whose other side is 2 ft in length

b A triangle whose hypotenuse is 13 in in length and whose other side is 8 in in length

26 For each of the given triangles, find the following, rounding your answers to two decimal places:

i Calculate the length of the hypotenuse **ii** Calculate the perimeter of the triangle

a A right triangle has two shorter sides measuring 15.4 in and 17.7 in

b A right isosceles triangle has two sides measuring 17 in

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27 Gisela asks her friend Chenzi if he can construct a right triangle with side lengths of 26 in, 10 in, and 24 in. Chenzi says it is not a right triangle. Look at Chenzi's work and describe the error.

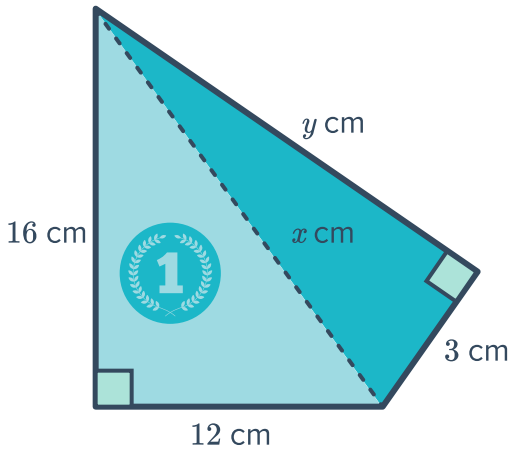
1	$26^2 + 10^2 = 676 + 100$	
2	$= 776$	
3	$24^2 = 576$	
4	$a^2 + b^2 = c^2$	Pythagorean theorem
5	$776 = 576$	

28 The diagonal of a square measures 15 units. Find the perimeter of the square in simplest radical form.

29 A sports association wants to redesign the trophy they award to the player of the season. The front view of one particular design is shown in the diagram:

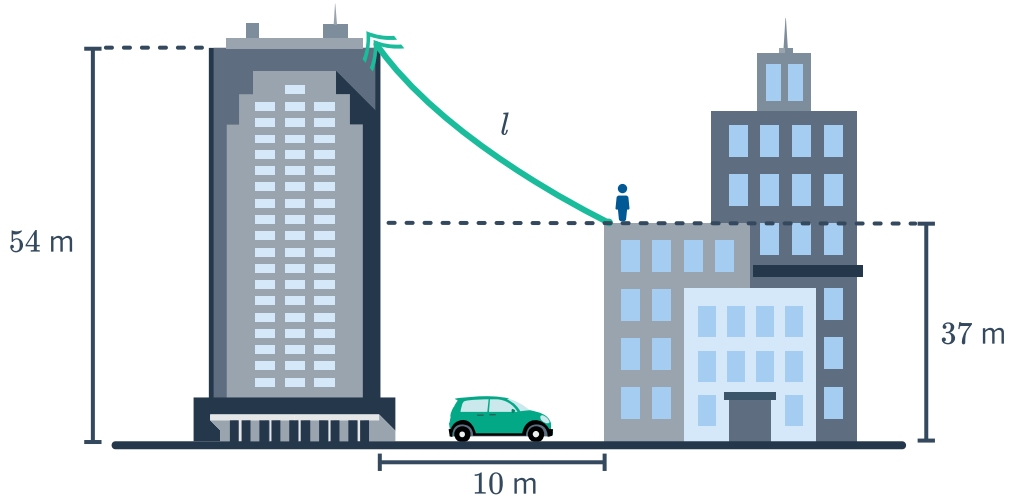
a Find the value of x

b Find the value of y, rounded to two decimal places.



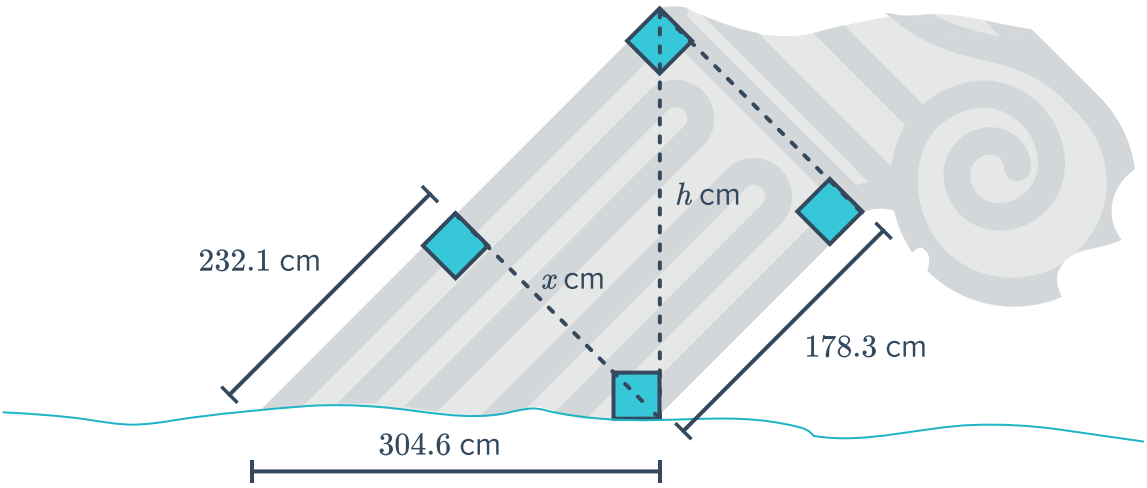
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30 A movie director wants to shoot a scene where the hero of the film fires a grappling hook from the roof of one building to the roof of another. The shorter building is $37\ \text{m}$ tall, the taller building is $54\ \text{m}$ tall and the street between them is $10\ \text{m}$ wide.



Find the minimum length of rope, l, needed for the grappling hook. Round your answer to two decimal places.

31 Archeologists have uncovered an ancient pillar which, after extensive digging, remains embedded in the ground. The lead researcher wants to record all of the dimensions of the pillar, including its height above the ground.
However, the team can only take certain measurements accurately without risking damage to the artifact. These measurements are shown in the diagram.

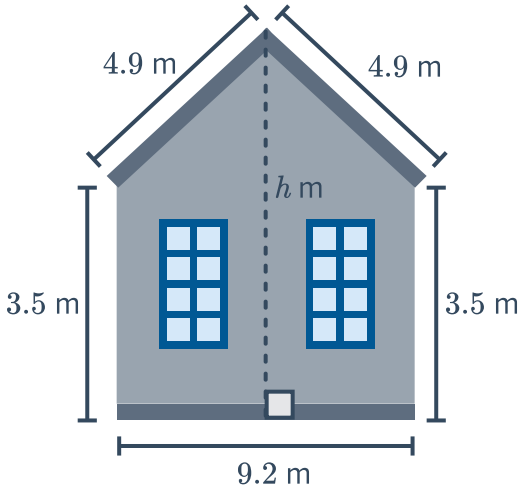


a Find the value of x, rounded to two decimal places.

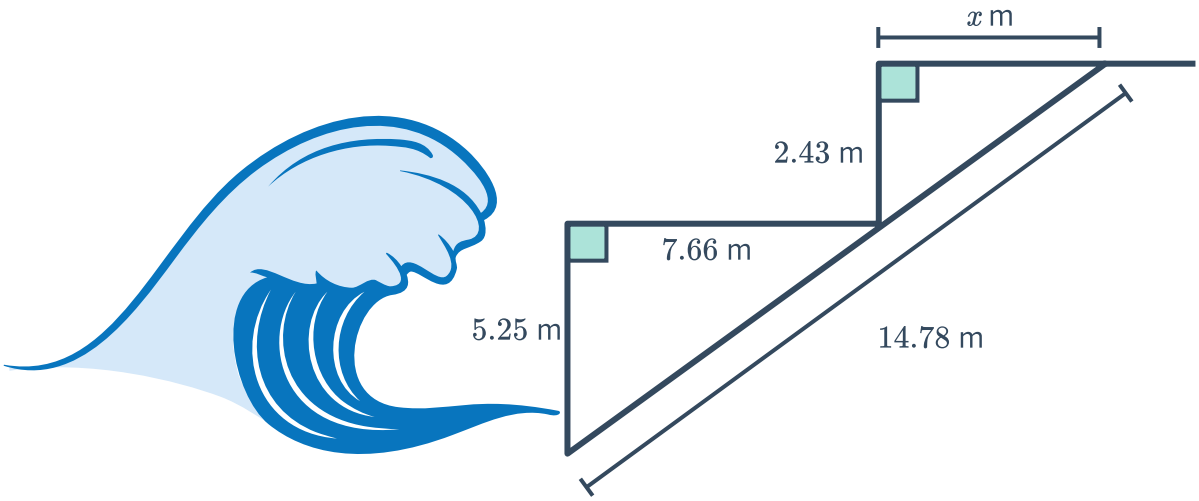
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b Find h, the height of the pillar, rounded to two decimal places.

32 Marge's house has the outer dimensions as shown in the diagram:
Find the height of the house, h, rounding your answer to one decimal place.

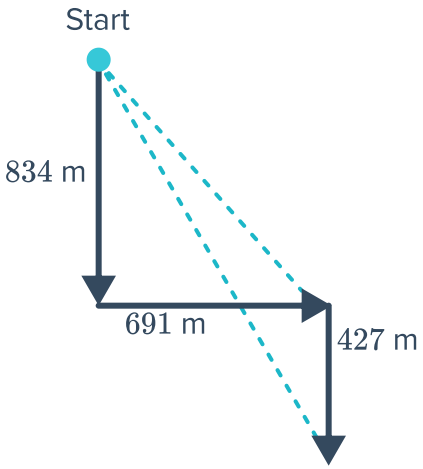


33 A city council plans to build a seawall and boardwalk along a local coastline. According to safety regulations, the seawall needs to be 5.25 m high and 7.66 m deep and will be built at the bottom of a 14.78 m long sloped section of shoreline. This means that the boardwalk will need to be built 2.43 m above the seawall, so that it is level with the public area near the beach.
Find the width of the boardwalk, x, rounding your answer to two decimal places.



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34 Angelia hikes south of her starting position for 834 ft and then 691 ft east, before stopping for a lunch break. She then travels south again for 427 ft before arriving at her final destination.



a Find the shortest distance between where Angelia started and where she stopped for lunch. Round your answer to two decimal places.

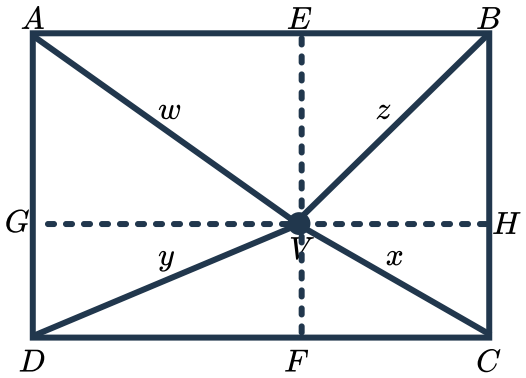
b Find the shortest distance between where Angelia started and where she finished her journey. Round your answer to two decimal places.

c Write a problem that uses the Pythagorean theorem and builds off of the diagram. Solve the problem.

35 Given:

- $ABCD$ is a rectangle
- $\overline{EF} \parallel \overline{AD}$
- $\overline{GH} \parallel \overline{DC}$

Prove: $w^2 + x^2 = y^2 + z^2$



36 A right triangle has a hypotenuse with a length of 65. If the lengths of the legs are integers, find two possible pairs of lengths that can complete the triangle. Explain your reasoning.

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